



MATH120: Applications of Derivatives

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Learning Objectives

- Find critical points by setting $f'(x) = 0$
- Classify intervals where f is increasing or decreasing
- Apply the First and Second Derivative Tests
- Solve optimization problems using calculus

Simplify each expression completely. Show all steps and circle your final answer.

Optimization

1. A company's profit function is $P(x) = -4x^2 + 10x + 47$, where x is units produced. Find the production level that maximizes profit.

$$-4x^2 + 10x + 47$$

Answer: _____

2. Revenue is $R(x) = -3x^2 + 21x + 18$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-3x^2 + 21x + 18$$

Answer: _____

3. A ball is thrown upward. Its height is $h(t) = -10t^2 + 53t + 8$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-10x^2 + 53x + 8$$

Answer: _____

4. A company's profit function is $P(x) = -4x^2 + 22x + 2$, where x is units produced. Find the production level that maximizes profit.

$$-4x^2 + 22x + 2$$

Answer: _____

5. Revenue is $R(x) = -2x^2 + 14x + 55$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-2x^2 + 14x + 55$$

Answer: _____



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6. A ball is thrown upward. Its height is $h(t) = -8t^2 + 40t + 2$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-8x^2 + 40x + 2$$

Answer: _____

7. A company's profit function is $P(x) = -1x^2 + 37x + 17$, where x is units produced. Find the production level that maximizes profit.

$$-1x^2 + 37x + 17$$

Answer: _____

8. Revenue is $R(x) = -3x^2 + 28x + 70$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-3x^2 + 28x + 70$$

Answer: _____

9. A ball is thrown upward. Its height is $h(t) = -12t^2 + 23t + 8$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-12x^2 + 23x + 8$$

Answer: _____

10. A company's profit function is $P(x) = -3x^2 + 24x + 35$, where x is units produced. Find the production level that maximizes profit.

$$-3x^2 + 24x + 35$$

Answer: _____

11. Revenue is $R(x) = -3x^2 + 18x + 19$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-3x^2 + 18x + 19$$

Answer: _____

12. A ball is thrown upward. Its height is $h(t) = -8t^2 + 20t + 7$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-8x^2 + 20x + 7$$

Answer: _____



13. A company's profit function is $P(x) = -3x^2 + 16x + 47$, where x is units produced. Find the production level that maximizes profit.

$$-3x^2 + 16x + 47$$

Answer: _____

14. Revenue is $R(x) = -3x^2 + 14x + 58$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-3x^2 + 14x + 58$$

Answer: _____

15. A ball is thrown upward. Its height is $h(t) = -12t^2 + 38t + 9$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-12x^2 + 38x + 9$$

Answer: _____

16. A company's profit function is $P(x) = -3x^2 + 12x + 15$, where x is units produced. Find the production level that maximizes profit.

$$-3x^2 + 12x + 15$$

Answer: _____

17. Revenue is $R(x) = -2x^2 + 26x + 37$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-2x^2 + 26x + 37$$

Answer: _____

18. A ball is thrown upward. Its height is $h(t) = -9t^2 + 27t + 10$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-9x^2 + 27x + 10$$

Answer: _____

19. A company's profit function is $P(x) = -4x^2 + 19x + 27$, where x is units produced. Find the production level that maximizes profit.

$$-4x^2 + 19x + 27$$

Answer: _____



20. Revenue is $R(x) = -3x^2 + 15x + 52$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-3x^2 + 15x + 52$$

Answer: _____

21. A ball is thrown upward. Its height is $h(t) = -9t^2 + 52t + 1$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-9x^2 + 52x + 1$$

Answer: _____

22. A company's profit function is $P(x) = -4x^2 + 19x + 44$, where x is units produced. Find the production level that maximizes profit.

$$-4x^2 + 19x + 44$$

Answer: _____

23. Revenue is $R(x) = -2x^2 + 28x + 79$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-2x^2 + 28x + 79$$

Answer: _____

Marginal analysis

24. Total cost is $C(x) = 2x^3 + -1x^2 + 10x + 115$. Find the marginal cost $MC = C'(1)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 2x^3 + -1x^2 + 10x + 115, x = 1$$

Answer: _____

25. Total cost is $C(x) = 1x^3 + -2x^2 + 9x + 78$. Find the marginal cost $MC = C'(2)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 1x^3 + -2x^2 + 9x + 78, x = 2$$

Answer: _____

26. Total cost is $C(x) = 1x^3 + -1x^2 + 11x + 173$. Find the marginal cost $MC = C'(3)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 1x^3 + -1x^2 + 11x + 173, x = 3$$

Answer: _____



27. Total cost is $C(x) = 1x^3 + -2x^2 + 12x + 54$. Find the marginal cost $MC = C'(1)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 1x^3 + -2x^2 + 12x + 54, x = 1$$

Answer: _____

28. Total cost is $C(x) = 2x^3 + -3x^2 + 11x + 105$. Find the marginal cost $MC = C'(2)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 2x^3 + -3x^2 + 11x + 105, x = 2$$

Answer: _____

29. Total cost is $C(x) = 2x^3 + -2x^2 + 7x + 181$. Find the marginal cost $MC = C'(1)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 2x^3 + -2x^2 + 7x + 181, x = 1$$

Answer: _____

30. Total cost is $C(x) = 1x^3 + -4x^2 + 15x + 92$. Find the marginal cost $MC = C'(1)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 1x^3 + -4x^2 + 15x + 92, x = 1$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Marginal analysis, Optimization. All answers verified by independent computation.

Solutions



Optimization

1. A company's profit function is $P(x) = -4x^2 + 10x + 47$, where x is units produced. Find the production level that maximizes profit.

$$-4x^2 + 10x + 47$$

→ Take the derivative: $P'(x) = -8x + 10$.

→ Set $P'(x) = 0$: $-8x + 10 = 0 \rightarrow x = 5/4$.

→ Since the leading coefficient is negative, $x = 5/4$ is a maximum. Max profit = $213/4$.

Answer: $x^* = -\frac{10}{-8} = \frac{5}{4}, f(x^*) = \frac{213}{4}$

2. Revenue is $R(x) = -3x^2 + 21x + 18$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-3x^2 + 21x + 18$$

→ Differentiate: $R'(x) = -6x + 21$.

→ Set equal to zero and solve: $x = 7/2$.

→ Maximum revenue = $R(7/2) = 219/4$.

Answer: $x^* = -\frac{21}{-6} = \frac{7}{2}, f(x^*) = \frac{219}{4}$

3. A ball is thrown upward. Its height is $h(t) = -10t^2 + 53t + 8$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-10x^2 + 53x + 8$$

→ $h'(t) = -20t + 53$. Set $h'(t) = 0$.

→ $t = 53/20$ seconds.

→ Maximum height = $h(53/20) = 3129/40$ feet.

Answer: $x^* = -\frac{53}{-20} = \frac{53}{20}, f(x^*) = \frac{3129}{40}$

4. A company's profit function is $P(x) = -4x^2 + 22x + 2$, where x is units produced. Find the production level that maximizes profit.

$$-4x^2 + 22x + 2$$

→ Take the derivative: $P'(x) = -8x + 22$.

→ Set $P'(x) = 0$: $-8x + 22 = 0 \rightarrow x = 11/4$.

→ Since the leading coefficient is negative, $x = 11/4$ is a maximum. Max profit = $129/4$.

Answer: $x^* = -\frac{22}{-8} = \frac{11}{4}, f(x^*) = \frac{129}{4}$

5. Revenue is $R(x) = -2x^2 + 14x + 55$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-2x^2 + 14x + 55$$

→ Differentiate: $R'(x) = -4x + 14$.

→ Set equal to zero and solve: $x = 7/2$.

→ Maximum revenue = $R(7/2) = 159/2$.

Answer: $x^* = -\frac{14}{-4} = \frac{7}{2}, f(x^*) = \frac{159}{2}$



6. A ball is thrown upward. Its height is $h(t) = -8t^2 + 40t + 2$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-8x^2 + 40x + 2$$

$$\rightarrow h'(t) = -16t + 40. \text{ Set } h'(t) = 0.$$

$$\rightarrow t = 5/2 \text{ seconds.}$$

$$\rightarrow \text{Maximum height} = h(5/2) = 52 \text{ feet.}$$

Answer: $x^* = -\frac{40}{-16} = \frac{5}{2}, f(x^*) = 52$

7. A company's profit function is $P(x) = -1x^2 + 37x + 17$, where x is units produced. Find the production level that maximizes profit.

$$-1x^2 + 37x + 17$$

$$\rightarrow \text{Take the derivative: } P'(x) = -2x + 37.$$

$$\rightarrow \text{Set } P'(x) = 0: -2x + 37 = 0 \rightarrow x = 37/2.$$

$$\rightarrow \text{Since the leading coefficient is negative, } x = 37/2 \text{ is a maximum. Max profit} = 1437/4.$$

Answer: $x^* = -\frac{37}{-2} = \frac{37}{2}, f(x^*) = \frac{1437}{4}$

8. Revenue is $R(x) = -3x^2 + 28x + 70$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-3x^2 + 28x + 70$$

$$\rightarrow \text{Differentiate: } R'(x) = -6x + 28.$$

$$\rightarrow \text{Set equal to zero and solve: } x = 14/3.$$

$$\rightarrow \text{Maximum revenue} = R(14/3) = 406/3.$$

Answer: $x^* = -\frac{28}{-6} = \frac{14}{3}, f(x^*) = \frac{406}{3}$

9. A ball is thrown upward. Its height is $h(t) = -12t^2 + 23t + 8$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-12x^2 + 23x + 8$$

$$\rightarrow h'(t) = -24t + 23. \text{ Set } h'(t) = 0.$$

$$\rightarrow t = 23/24 \text{ seconds.}$$

$$\rightarrow \text{Maximum height} = h(23/24) = 913/48 \text{ feet.}$$

Answer: $x^* = -\frac{23}{-24} = \frac{23}{24}, f(x^*) = \frac{913}{48}$

10. A company's profit function is $P(x) = -3x^2 + 24x + 35$, where x is units produced. Find the production level that maximizes profit.

$$-3x^2 + 24x + 35$$

$$\rightarrow \text{Take the derivative: } P'(x) = -6x + 24.$$

$$\rightarrow \text{Set } P'(x) = 0: -6x + 24 = 0 \rightarrow x = 4.$$

$$\rightarrow \text{Since the leading coefficient is negative, } x = 4 \text{ is a maximum. Max profit} = 83.$$

Answer: $x^* = -\frac{24}{-6} = 4, f(x^*) = 83$



11. Revenue is $R(x) = -3x^2 + 18x + 19$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-3x^2 + 18x + 19$$

→ Differentiate: $R'(x) = -6x + 18$.

→ Set equal to zero and solve: $x = 3$.

→ Maximum revenue = $R(3) = 46$.

Answer: $x^* = -\frac{18}{-6} = 3, f(x^*) = 46$

12. A ball is thrown upward. Its height is $h(t) = -8t^2 + 20t + 7$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-8x^2 + 20x + 7$$

→ $h'(t) = -16t + 20$. Set $h'(t) = 0$.

→ $t = 5/4$ seconds.

→ Maximum height = $h(5/4) = 39/2$ feet.

Answer: $x^* = -\frac{20}{-16} = \frac{5}{4}, f(x^*) = \frac{39}{2}$

13. A company's profit function is $P(x) = -3x^2 + 16x + 47$, where x is units produced. Find the production level that maximizes profit.

$$-3x^2 + 16x + 47$$

→ Take the derivative: $P'(x) = -6x + 16$.

→ Set $P'(x) = 0$: $-6x + 16 = 0 \rightarrow x = 8/3$.

→ Since the leading coefficient is negative, $x = 8/3$ is a maximum. Max profit = $205/3$.

Answer: $x^* = -\frac{16}{-6} = \frac{8}{3}, f(x^*) = \frac{205}{3}$

14. Revenue is $R(x) = -3x^2 + 14x + 58$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-3x^2 + 14x + 58$$

→ Differentiate: $R'(x) = -6x + 14$.

→ Set equal to zero and solve: $x = 7/3$.

→ Maximum revenue = $R(7/3) = 223/3$.

Answer: $x^* = -\frac{14}{-6} = \frac{7}{3}, f(x^*) = \frac{223}{3}$

15. A ball is thrown upward. Its height is $h(t) = -12t^2 + 38t + 9$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-12x^2 + 38x + 9$$

→ $h'(t) = -24t + 38$. Set $h'(t) = 0$.

→ $t = 19/12$ seconds.

→ Maximum height = $h(19/12) = 469/12$ feet.

Answer: $x^* = -\frac{38}{-24} = \frac{19}{12}, f(x^*) = \frac{469}{12}$



16. A company's profit function is $P(x) = -3x^2 + 12x + 15$, where x is units produced. Find the production level that maximizes profit.

$$-3x^2 + 12x + 15$$

→ Take the derivative: $P'(x) = -6x + 12$.

→ Set $P'(x) = 0$: $-6x + 12 = 0 \rightarrow x = 2$.

→ Since the leading coefficient is negative, $x = 2$ is a maximum. Max profit = 27.

Answer: $x^* = -\frac{12}{-6} = 2, f(x^*) = 27$

17. Revenue is $R(x) = -2x^2 + 26x + 37$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-2x^2 + 26x + 37$$

→ Differentiate: $R'(x) = -4x + 26$.

→ Set equal to zero and solve: $x = 13/2$.

→ Maximum revenue = $R(13/2) = 243/2$.

Answer: $x^* = -\frac{26}{-4} = \frac{13}{2}, f(x^*) = \frac{243}{2}$

18. A ball is thrown upward. Its height is $h(t) = -9t^2 + 27t + 10$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-9x^2 + 27x + 10$$

→ $h'(t) = -18t + 27$. Set $h'(t) = 0$.

→ $t = 3/2$ seconds.

→ Maximum height = $h(3/2) = 121/4$ feet.

Answer: $x^* = -\frac{27}{-18} = \frac{3}{2}, f(x^*) = \frac{121}{4}$

19. A company's profit function is $P(x) = -4x^2 + 19x + 27$, where x is units produced. Find the production level that maximizes profit.

$$-4x^2 + 19x + 27$$

→ Take the derivative: $P'(x) = -8x + 19$.

→ Set $P'(x) = 0$: $-8x + 19 = 0 \rightarrow x = 19/8$.

→ Since the leading coefficient is negative, $x = 19/8$ is a maximum. Max profit = $793/16$.

Answer: $x^* = -\frac{19}{-8} = \frac{19}{8}, f(x^*) = \frac{793}{16}$

20. Revenue is $R(x) = -3x^2 + 15x + 52$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-3x^2 + 15x + 52$$

→ Differentiate: $R'(x) = -6x + 15$.

→ Set equal to zero and solve: $x = 5/2$.

→ Maximum revenue = $R(5/2) = 283/4$.

Answer: $x^* = -\frac{15}{-6} = \frac{5}{2}, f(x^*) = \frac{283}{4}$



21. A ball is thrown upward. Its height is $h(t) = -9t^2 + 52t + 1$ feet. Find the time t at which it reaches maximum height, then find that maximum height.

$$-9x^2 + 52x + 1$$

$$\rightarrow h'(t) = -18t + 52. \text{ Set } h'(t) = 0.$$

$$\rightarrow t = 26/9 \text{ seconds.}$$

$$\rightarrow \text{Maximum height} = h(26/9) = 685/9 \text{ feet.}$$

Answer: $x^* = -\frac{52}{-18} = \frac{26}{9}, f(x^*) = \frac{685}{9}$

22. A company's profit function is $P(x) = -4x^2 + 19x + 44$, where x is units produced. Find the production level that maximizes profit.

$$-4x^2 + 19x + 44$$

$$\rightarrow \text{Take the derivative: } P'(x) = -8x + 19.$$

$$\rightarrow \text{Set } P'(x) = 0: -8x + 19 = 0 \rightarrow x = 19/8.$$

$$\rightarrow \text{Since the leading coefficient is negative, } x = 19/8 \text{ is a maximum. Max profit} = 1065/16.$$

Answer: $x^* = -\frac{19}{-8} = \frac{19}{8}, f(x^*) = \frac{1065}{16}$

23. Revenue is $R(x) = -2x^2 + 28x + 79$. Find the number of units x that maximizes revenue, and compute that maximum revenue.

$$-2x^2 + 28x + 79$$

$$\rightarrow \text{Differentiate: } R'(x) = -4x + 28.$$

$$\rightarrow \text{Set equal to zero and solve: } x = 7.$$

$$\rightarrow \text{Maximum revenue} = R(7) = 177.$$

Answer: $x^* = -\frac{28}{-4} = 7, f(x^*) = 177$



Marginal analysis

24. Total cost is $C(x) = 2x^3 + -1x^2 + 10x + 115$. Find the marginal cost $MC = C'(1)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 2x^3 + -1x^2 + 10x + 115, x = 1$$

$$\rightarrow C'(x) = 6x^2 + -2x + 10.$$

$$\rightarrow MC = C'(1) = 14.$$

$\rightarrow$ Since $MC > 0$, cost is increasing at $x = 1$.

Answer: $f'(x) = 6x^2 + -2x + 10, f'(1) = 14$

25. Total cost is $C(x) = 1x^3 + -2x^2 + 9x + 78$. Find the marginal cost $MC = C'(2)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 1x^3 + -2x^2 + 9x + 78, x = 2$$

$$\rightarrow C'(x) = 3x^2 + -4x + 9.$$

$$\rightarrow MC = C'(2) = 13.$$

$\rightarrow$ Since $MC > 0$, cost is increasing at $x = 2$.

Answer: $f'(x) = 3x^2 + -4x + 9, f'(2) = 13$

26. Total cost is $C(x) = 1x^3 + -1x^2 + 11x + 173$. Find the marginal cost $MC = C'(3)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 1x^3 + -1x^2 + 11x + 173, x = 3$$

$$\rightarrow C'(x) = 3x^2 + -2x + 11.$$

$$\rightarrow MC = C'(3) = 32.$$

$\rightarrow$ Since $MC > 0$, cost is increasing at $x = 3$.

Answer: $f'(x) = 3x^2 + -2x + 11, f'(3) = 32$

27. Total cost is $C(x) = 1x^3 + -2x^2 + 12x + 54$. Find the marginal cost $MC = C'(1)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 1x^3 + -2x^2 + 12x + 54, x = 1$$

$$\rightarrow C'(x) = 3x^2 + -4x + 12.$$

$$\rightarrow MC = C'(1) = 11.$$

$\rightarrow$ Since $MC > 0$, cost is increasing at $x = 1$.

Answer: $f'(x) = 3x^2 + -4x + 12, f'(1) = 11$

28. Total cost is $C(x) = 2x^3 + -3x^2 + 11x + 105$. Find the marginal cost $MC = C'(2)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 2x^3 + -3x^2 + 11x + 105, x = 2$$

$$\rightarrow C'(x) = 6x^2 + -6x + 11.$$

$$\rightarrow MC = C'(2) = 23.$$

$\rightarrow$ Since $MC > 0$, cost is increasing at $x = 2$.

Answer: $f'(x) = 6x^2 + -6x + 11, f'(2) = 23$



29. Total cost is $C(x) = 2x^3 + -2x^2 + 7x + 181$. Find the marginal cost $MC = C'(1)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 2x^3 + -2x^2 + 7x + 181, \quad x = 1$$

$$\rightarrow C'(x) = 6x^2 + -4x + 7.$$

$$\rightarrow MC = C'(1) = 9.$$

$\rightarrow$ Since $MC > 0$, cost is increasing at $x = 1$.

Answer: $f'(x) = 6x^2 + -4x + 7, \quad f'(1) = 9$

30. Total cost is $C(x) = 1x^3 + -4x^2 + 15x + 92$. Find the marginal cost $MC = C'(1)$, and determine whether cost is increasing or decreasing at that production level.

$$f(x) = 1x^3 + -4x^2 + 15x + 92, \quad x = 1$$

$$\rightarrow C'(x) = 3x^2 + -8x + 15.$$

$$\rightarrow MC = C'(1) = 10.$$

$\rightarrow$ Since $MC > 0$, cost is increasing at $x = 1$.

Answer: $f'(x) = 3x^2 + -8x + 15, \quad f'(1) = 10$

